

Titanic Exploratory Data Analysis

Objective:

Identify patterns and anomalies linked to survival.

Dataset size:

891 rows, 12 columns

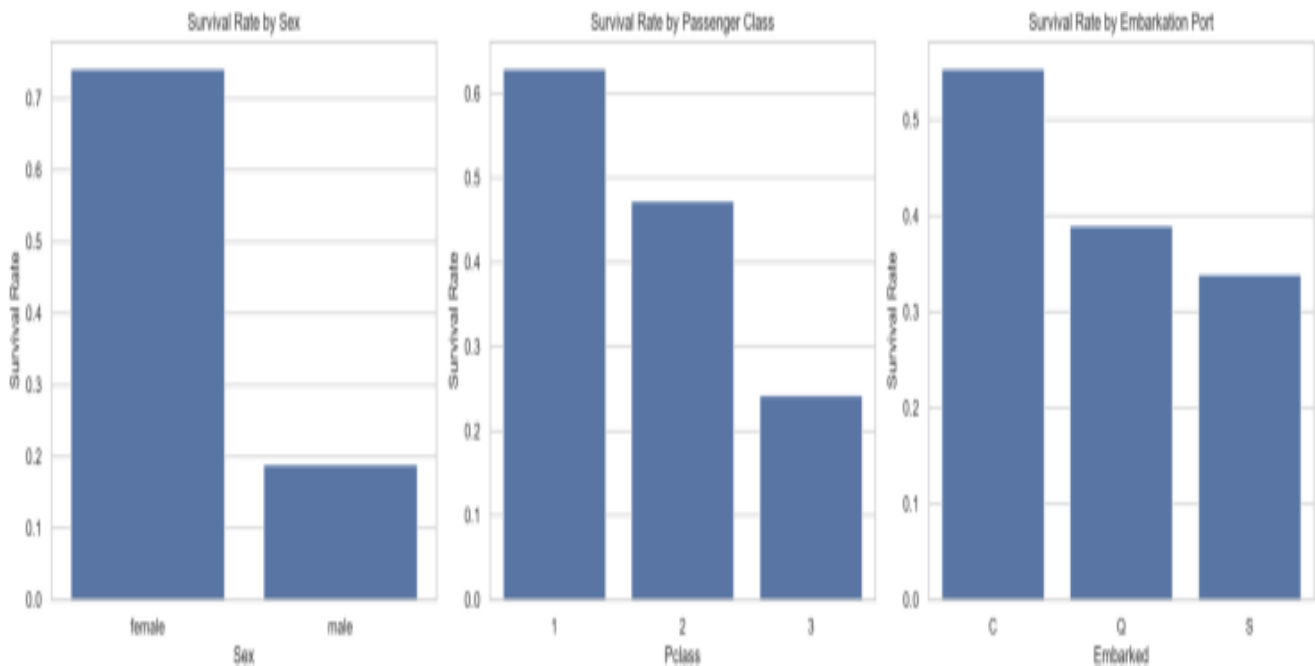
Survival rate:

38.38%

Key findings:

- Women had much higher survival than men
- First-class passengers survived more often
- Higher fares corresponded to better survival outcomes
- Age and family size showed secondary influence

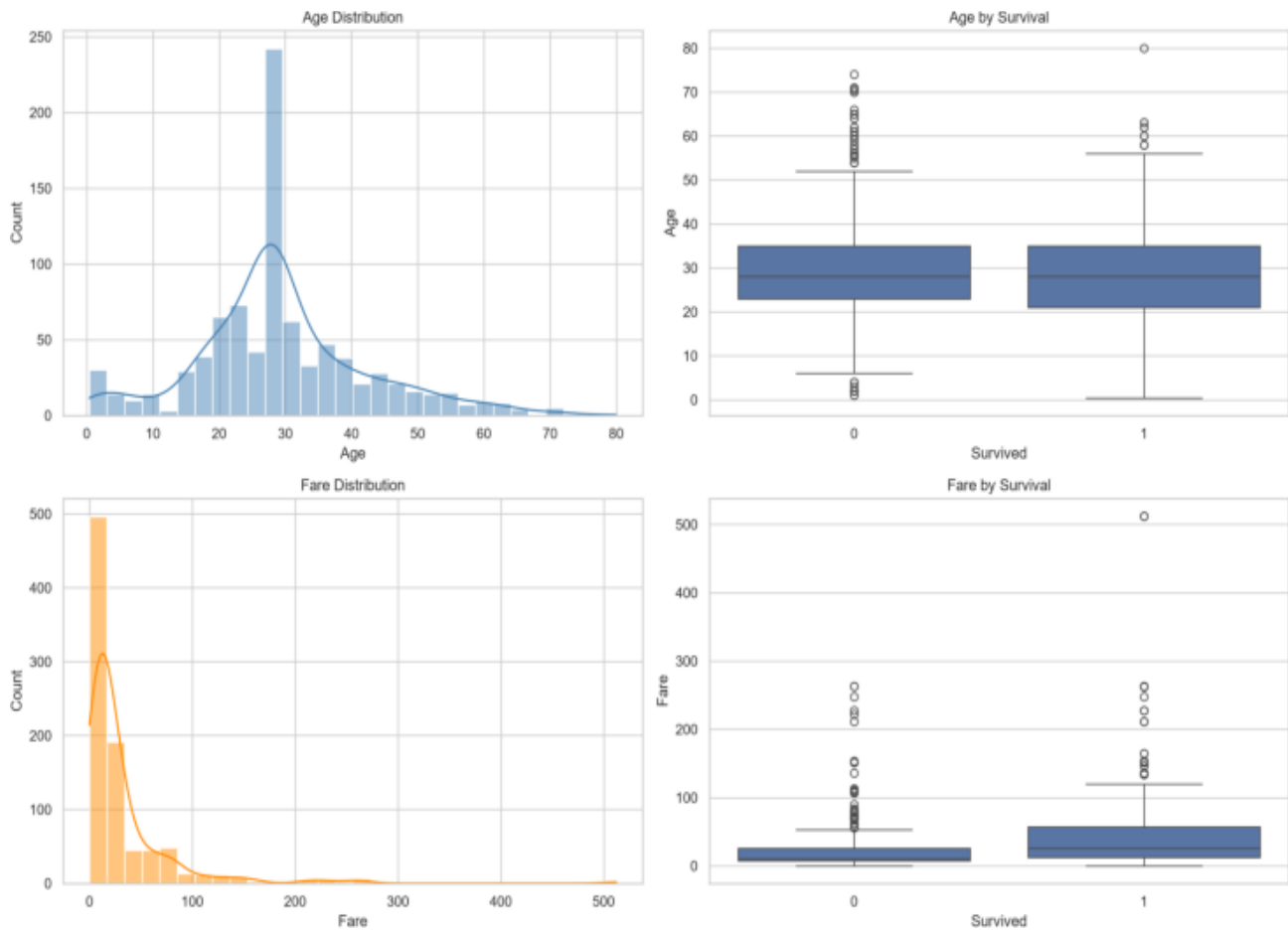
Survival Patterns by Group:



- Sex: female passengers had a much higher survival rate than male passengers
- Passenger class: first-class passengers had the highest survival rate, while third-class passengers had the lowest
- Embarkation port: survival varied by boarding port, with higher survival seen for some groups

Interpretation: Sex and passenger class are the clearest signals in the data.

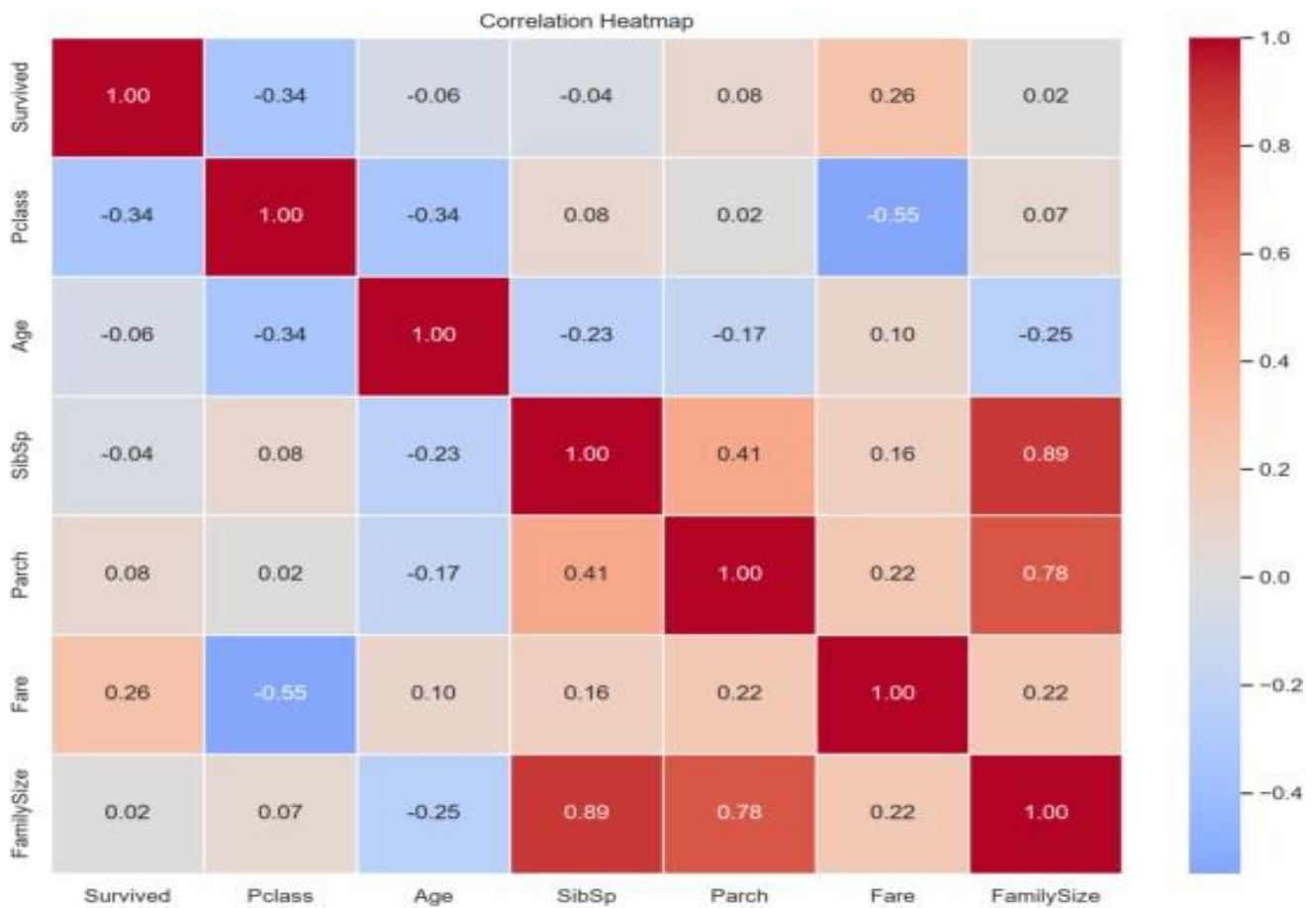
Distribution and Risk Signals:



- Age distribution shows a large concentration of passengers between 20 and 40 years
- Survivors tended to be younger on average than non-survivors
- Fare distribution was highly skewed, with a small number of very high-fare tickets
- Passengers who paid higher fares were more likely to survive

Interpretation: Age and fare distributions show that wealth and cabin class greatly influenced survival odds.

Correlation and Pairwise Relationships:



- Passenger class, fare, and sex show the strongest relationships with survival.
- Family size and age had moderate influence.
- The data suggests that social and economic factors strongly affected survival likelihood.

Summary:

- Sex was the strongest factor in survival.
- Higher-class passengers had better survival odds.
- Fare acted as a proxy for social and economic status.
- Age was a secondary contributor to survival.
- Missing values in Age and Cabin should be handled carefully in modelling.